

Password Best Practices -- Guided Notes

Unit tp-6.13 | CompTIA Network+ N10-009 Obj. FC0-U71 6.3 | N10-008 FC0-U71 6.3

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

A strong password should be: at least _____ characters long, contain a mix of _____, _____, _____, and _____, and not include obvious personal information like _____, _____, or pet names. Length is the single most important factor -- a 16-character passphrase beats a 10-character complex password in brute force resistance.

Question 2

Password _____ means using the same password on multiple sites. It is extremely dangerous because if any one site is breached and your password is exposed, attackers use _____ stuffing attacks to automatically try that password on banking, email, and other accounts. The fix is to use a unique password for _____ account.

Question 3

A password _____ stores all your passwords in an encrypted vault, protected by a single strong _____ password. It can generate and autofill _____, unique passwords for every site. Examples include Bitwarden, 1Password, and LastPass. The browser saves them for _____ of use. This is the recommended solution to the reuse problem.

Question 4

Passwords should never be _____ insecurely (written on sticky notes, sent in plain text email, stored in an unencrypted text file). Never share passwords via _____ or messaging apps. If you must write a master password down as a backup, store it in a physically _____ location (safe, locked drawer -- not a monitor sticky note).

Question 5

Default passwords on devices (routers, printers, IP cameras) must be _____ immediately on setup. Many devices ship with the same default credentials (like admin/admin or admin/password) that are _____ documented online. Attackers scan for devices with _____ credentials as an automated first step. Failure to change defaults is one of the most common _____ vulnerabilities in home and

small business networks.

Question 6

Password _____ policies require users to change passwords on a regular schedule (e.g., every 90 days). NIST now recommends _____ periodic rotation unless there is evidence of compromise, because forced rotation leads to _____ password choices (e.g., Password1, Password2). Instead, NIST recommends checking passwords against known _____ lists at creation.

Question 7

Passwords in databases must be stored as _____ not plaintext. A _____ function converts the password to a fixed-length digest. A random value called a _____ is added before hashing to ensure two users with the same password have different stored hashes, defeating _____ table attacks.

Question 8 -- Real World Application

REAL WORLD: A student uses the same password ('Gopackers88!') for their school email, Instagram, a gaming forum, and their bank. The gaming forum is breached and their email/password is posted publicly. Within 24 hours, their Instagram is taken over and their bank issues a fraud alert. Explain step by step how the attacker moved from a gaming forum breach to a bank account attempt.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-6-13-password-best-practices/tp-6-13-password-best-practices-answer-key.pdf
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